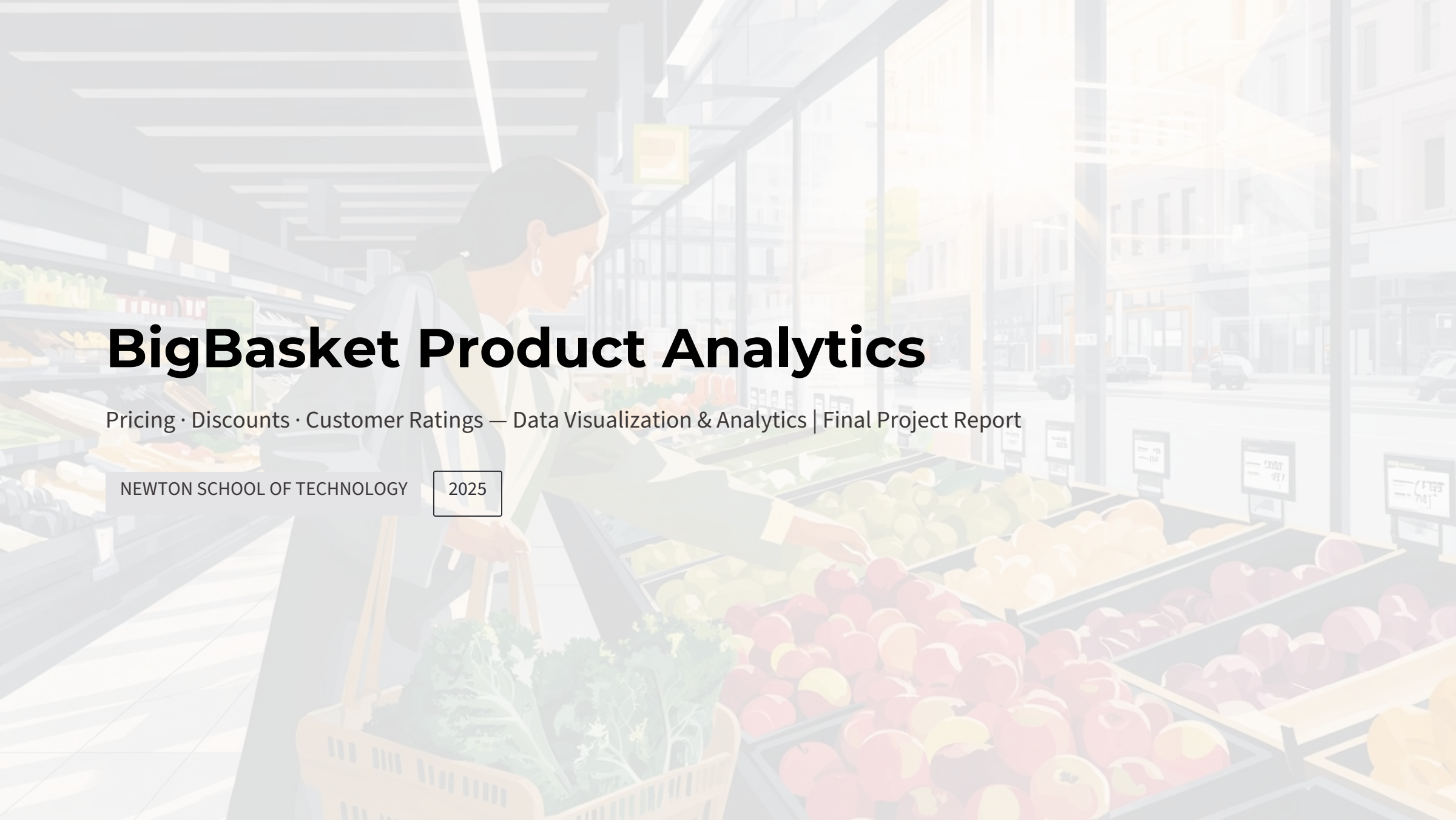




BigBasket Product Analytics

Pricing · Discounts · Customer Ratings — Data Visualization & Analytics | Final Project Report

NEWTON SCHOOL OF TECHNOLOGY

2025

Project Overview

Project Details

Sector: E-Commerce / Online Grocery Retail

Domain: India — BigBasket Platform

Dataset: Kaggle — 23,000+ Product Listings

Tech Stack: Python · Tableau · GitHub

Submission: 2025

GitHub: github.com/amithere/capstone2

Tableau: public.tableau.com/app/profile/aneesh.amiti/viz/

Our Team

Amiti Aneesh

Data Cleaning & ETL Pipeline

Lakshith

Exploratory & Statistical Analysis

Chanakya Shinde

Tableau Dashboard & Visualization

Sai Praneeth Sharma

Business Insights & Final Report

How can BigBasket attract more customers? Analysing pricing, discount patterns and customer ratings to deliver actionable strategy.

Executive Summary

BigBasket, India's largest online grocery platform, faces rising competition from Blinkit, Zepto, and JioMart. This project analyses **23,000+ product listings** to answer: what drives pricing, discounts, and customer ratings — and how can these insights improve strategy, assortment, and satisfaction? Using a Python ETL pipeline and 10 structured analyses (EDA + statistical tests), the team extracted the following key findings.

Finding 1

Sale price–rating correlation is **very weak** — price alone does not drive satisfaction.

Finding 2

Very heavily discounted products trend toward **lower ratings**, signalling quality issues rather than value.

Finding 3

Budget products ($\leq ₹100$) dominate volume but carry the **lowest ratings** — the biggest quality drag on the platform.

Finding 4

Mid-range products (₹100–₹500) achieve the **best satisfaction-to-volume balance** and should lead promotional strategy.

Sector Context & Problem Statement

Business Context

India's online grocery sector is growing rapidly, driven by smartphone penetration, urbanisation, and post-pandemic behavioural change. BigBasket (Tata Group, est. 2011) is the market leader across 30+ cities, but faces structural pressure from thin margins, complex perishable logistics, and quick-commerce rivals.

This analysis serves BigBasket's **Category Management, Pricing Strategy, and Marketing teams**. Pricing and discounting directly affect customer acquisition cost, gross margin, and retention.

Formal Research Question

What are the key drivers of product pricing, discount patterns, and customer ratings across categories and brands on BigBasket — and how can these insights optimise pricing strategy, improve assortment, and boost satisfaction?

Scope Definition

In Scope	Out of Scope
23,000+ product listings — all categories and brands	Real-time or transactional sales volume data
Pricing trends, discounts, price segmentation	Customer-level demographic/purchase history
Rating analysis vs pricing and brand	Supply chain, logistics, or cost-side analysis
Brand & category performance benchmarking	Machine learning / predictive modelling
Correlation, regression, ANOVA across 10 analyses	Time-series trend analysis (no time dimension in data)

Data Description & ETL Pipeline

Source:Kaggle —BigBasketEntireProductList(surajjha101).Astaticcatalogsnapshotof23,000+ product listings across all major BigBasket categories.

Column	Type	Description
product	Text	Full product title as listed on BigBasket
category	Categorical	Broad grouping (~13 unique categories)
sub_category	Categorical	Fine-grained grouping within each category (100+)
brand	Categorical	Brand name — 1,000+ unique values
sale_price	Float (₹)	Price sold on BigBasket; converted from string
market_price	Float (₹)	Listed MRP; may differ from true market price
rating	Float 0–5	Consumer rating; missing for some products
discount_amount ♦	Float (₹)	Engineered: market_price – sale_price
discount_pct ♦	Float (%)	Engineered: (discount_amount / market_price) × 100
price_segment ♦	Categorical	Engineered: Budget ≤₹100 Mid-range ₹100–500 Premium >₹500
is_discounted ♦	Boolean	Engineered: True if sale_price < market_price
rating_segment ♦	Categorical	Engineered: Low <3 Medium 3–4 High >4

 ♦ = feature engineered during ETL. Data limitations: static snapshot only; no sales volume; missing ratings biased toward newer products; MRP may not reflect competitor prices.

ETL Cleaning Steps

Issue Found	Columns Affected	Treatment
Missing values	rating, brand, description	Imputed/flagged; description excluded from analysis
Prices as strings with ₹ symbol	sale_price, market_price	Stripped symbol; converted to float64
Inconsistent brand name casing	brand	Standardised to title case
sale_price > market_price anomalies	Both price cols	Flagged; excluded from discount analysis
Duplicate product entries	All columns	Identified and removed
Outliers in prices	sale_price, market_price	IQR detection; retained with flags

Exploratory Data Analysis — 10 Analyses

Reference: `notebooks/03_eda.ipynb`. All 10 analyses performed using `matplotlib` and `seaborn`.

1	Descriptive Statistics Right-skewed price distribution (mean > median) confirms most products are low-to-mid priced. Rating mean and discount share establish KPI baselines.
2	Category Distribution Long-tail distribution — top 3 categories dominate listings. Most categories are thinly stocked, representing untapped assortment expansion opportunities.
3	Top Brands Analysis A small number of brands dominate catalog listings. High concentration creates co-marketing opportunity but also supply-side dependency risk.
4	Price Distribution Strong right skew. Budget and Mid-range dominate volume; Premium is a long tail. BigBasket operates as a value-oriented platform.
5	Rating Distribution Ratings cluster in 3.5–4.5 band — moderately positive perception overall. Low-rated segment (<3.0) is a reputational risk.
1	Discount Analysis Near-universal discounting normalises markdown expectations. Rating vs discount scatter shows no upward trend — heavy discounts do not earn better ratings.
2	Price vs Rating Correlation Correlation classified as very weak. No meaningful linear relationship between price and satisfaction. Higher price cannot be justified by rating uplift.
3	Most Expensive Categories Premium pricing is concentrated in specific categories. These require different discount depth and promotion policy compared to mass-market staples.
4	Brand Performance by Rating Significant spread between top and bottom performers. Top-rated brands are candidates for featured placement; bottom-rated brands should be reviewed.
5	Price Segments Deep Dive Budget is volume-heavy but quality-poor. Mid-range is the satisfaction sweet spot. Premium underdelivers its price promise. Each segment needs independent strategy.

Tableau Dashboard — Price, Discount & Trust

BigBasketProductIntelligence —interactivedashboardsurfacingKPIs,segmentcomparisons,andthediscountvs.rating quadrantforcategorymanagers andleadership.

56.9%

DiscountCoverage
of SKUs are discounted — primary pricing lever

22.0%

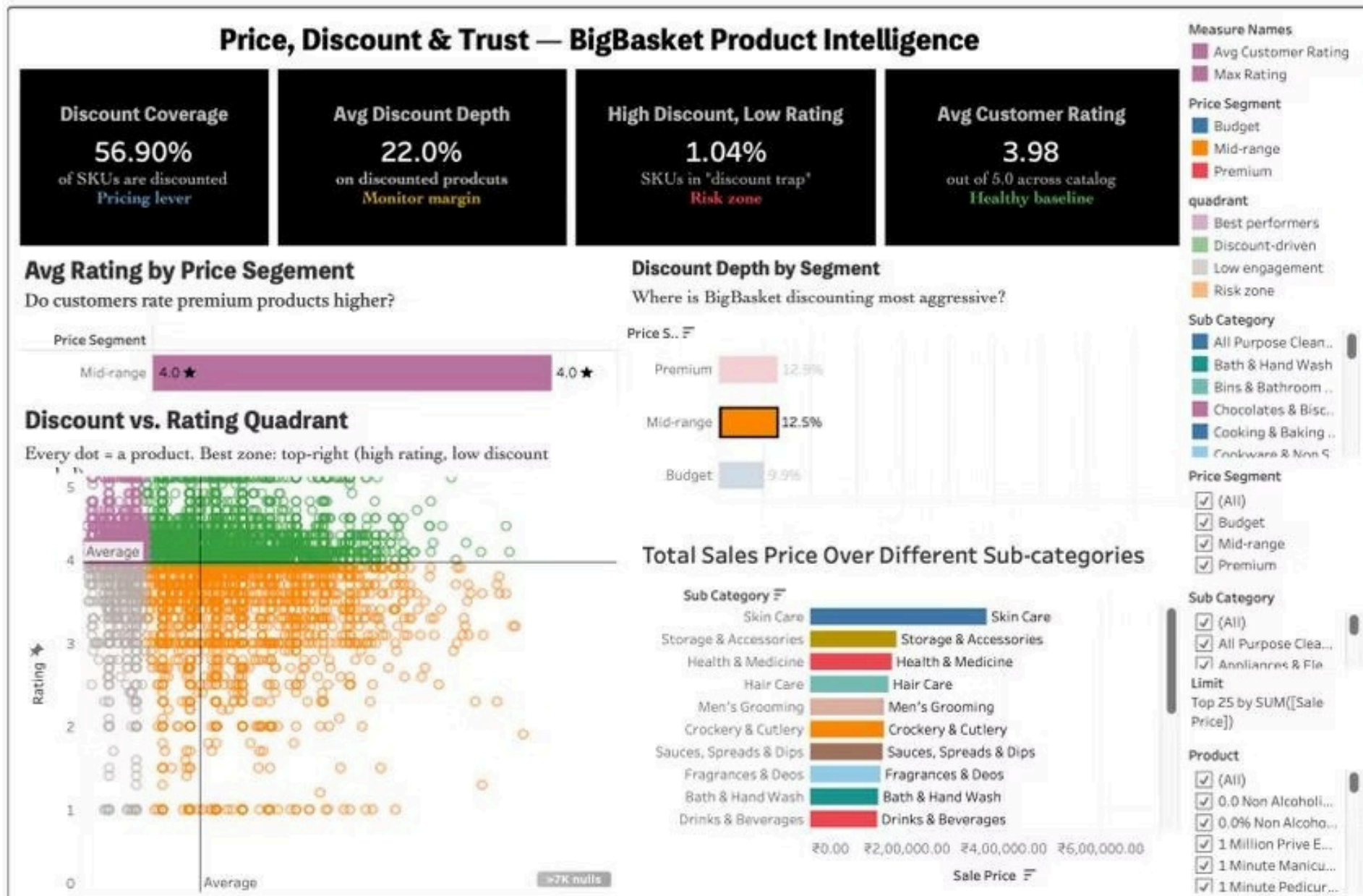
AvgDiscountDepth
on discounted products — monitor margin

1.04%

DiscountTrapSKUs
High discount, low rating — risk zone

3.98

out of 5.0 across catalog — healthy baseline



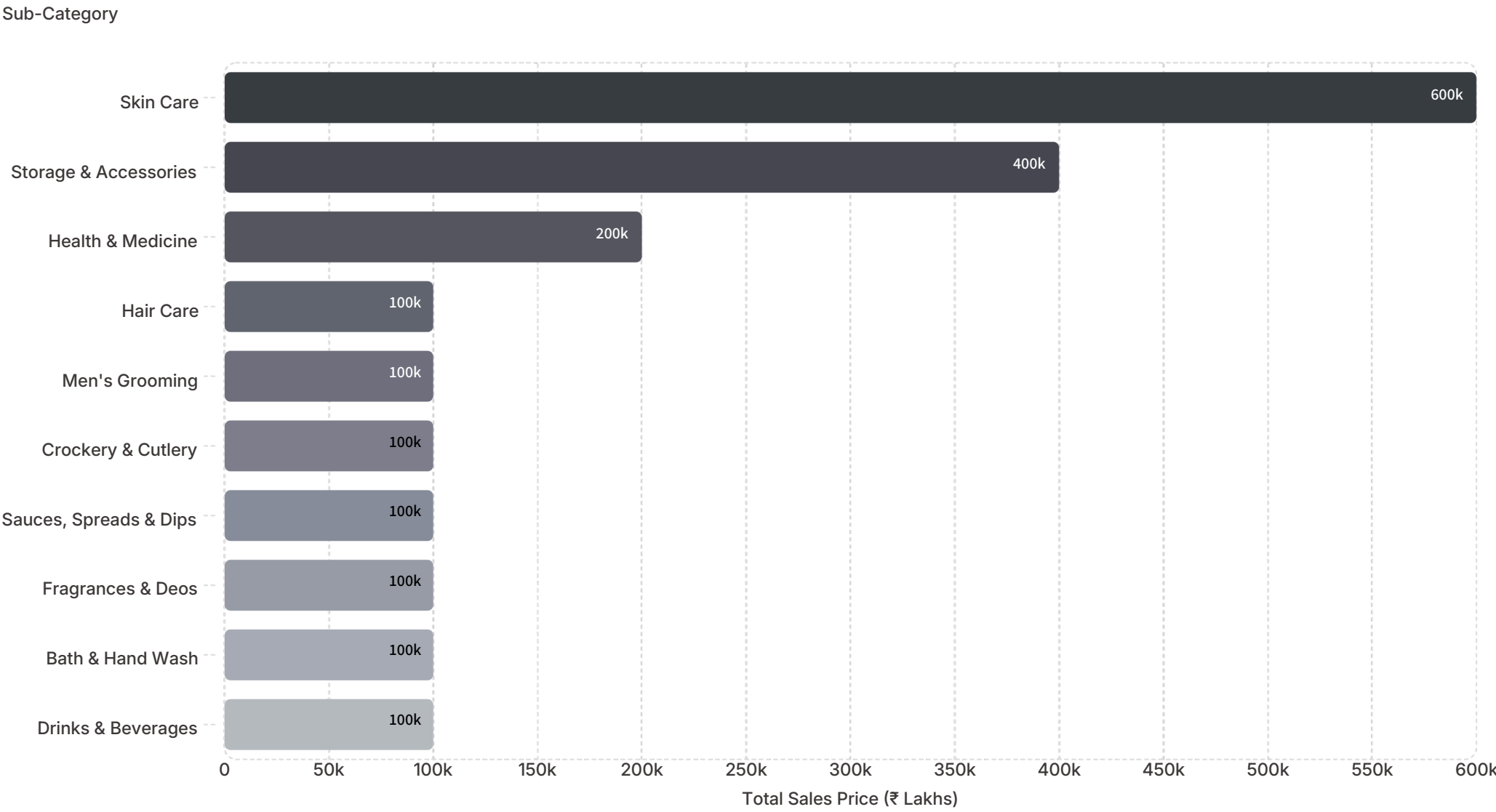
Avg Rating by Price Segment

Price Segment	Avg Rating
Mid-range	4.0
Premium	Moderate
Budget	Lowest

Discount Depth by Segment

Price Segment	Discount Depth
Premium	12.5%
Mid-range	12.5%
Budget	0.0%

Top Sub-categories by Total Sales Price



i Interactive Filters: Multi-select category filter · Price segment (Budget/Mid-range/Premium) · Rating segment (Low/Medium/High) · Hover tooltips on all charts. View dashboard: [Tableau Public](#)

Insights & Recommendations



Budget Quality Drag

Budgetproducts(≤₹100)dominate volume but have the lowest average ratings — the largest quality drag on the platform. **Action:** Curate aggressively; remove/improve lowest-rated Budget SKUs.



Price ≠ Satisfaction

Price–rating correlation isvery weak; customers do not reward higher prices with better ratings. **Action:** Decouple pricing decisions from rating-uplift expectations.



Mid-Range Sweet Spot

Mid-rangeproducts(₹100–₹500) achieve the highest average ratings — the platform's quality sweet spot. **Action:** Prioritise this segment in homepage and marketing promotion.



Discount Trap Risk

Veryheavilydiscountedproducts trend toward lower ratings — deep discounts signal poor quality, not value. **Action:** Audit all products with discount_pct >40% AND rating <3.5.

Full Recommendations Matrix

Prio rity	Recommendation	Insight Basis	Expected Impact	Feasibility
1 — High	Audit and delist deeply discounted, low-rated SKUs (discount_pct >40%, rating <3.5)	Insights 4, 2	Platform avg rating ↑; reduced dissatisfied customers; better discount ROI	Medium
2 — High	Promote mid-range, high-rating brands (avg rating >4.0, ≥10 products) on homepage, email, and app campaigns	Insights 3, 5	CTR on featured products ↑; stronger brand partnerships; improved conversion	High
3 — Medi um	Content audit in sub-categories with avg rating <3.5: standardise product descriptions	Insights 6, 8	Return rate ↓; post-purchase satisfaction ↑; better organic search	Medium
4 — Medi um	Replace always-on small discounts with structured Sale Events tied to festivals/seasons	Insight 7	Purchase urgency ↑; gross margin ↑ in non-event periods; stronger perceived value	Med-High
5 — Medi um	Launch 'Best Value' badge for Budget products with rating >4.0; removal pipeline for Budget SKUs with rating <3.0	Insight 1	Budget segment avg rating ↑; price-conscious customer satisfaction ↑	High

Impact Estimation & Limitations

Estimated Impact by Recommendation

Recommendation	Target Metric	Estimated Impact
Audit low-rated / high-discount SKUs	Platform avg rating	↑ 0.1–0.2 pts
Mid-range brand promotion	Marketing CTR	↑ 15–25%
Product description improvement	Return rate	↓ 5–10%
Targeted promotion events	Gross margin (non-event)	↑ 2–4%
Budget Quality Badge + removal	Budget segment avg rating	↑ 0.2–0.3 pts

 Urgency: Quick-commerce rivals are actively investing in catalog quality. First-mover advantage in assortment curation will be a key differentiator in the next 12–18 months.

Study Limitations

- ### No Sales Volume Data

All products treated equally regardless of purchase frequency; a 1-review product equals a bestseller.
- ### Static Snapshot

Directional pricing or rating trends cannot be identified; findings describe one point in time.
- ### Missing Rating Bias

If lower-quality products are less likely to be rated, observed average ratings are optimistically biased upward.
- ### Correlation ≠ Causation

Statistical associations found; we cannot conclude that reducing discounts will cause ratings to rise.
- ### Low Regression R²

Price features explain little rating variance — missing variables (description quality, brand age) not in dataset.

Statistical Tests Summary

Test	Variables	Finding	Conclusion
Pearson Correlation	sale_price vs rating	Very weak (≈ 0)	Price does not predict rating
Pearson Correlation	discount_pct vs rating	Weak negative	Higher discounts → lower ratings
Group Comparison	High vs low discount groups	No positive difference in rating	Discounts don't improve satisfaction
One-way ANOVA	Rating across categories	Statistically significant difference	Category drives rating; set category-level targets
Linear Regression	Price features → rating	Low R-squared	Price features poorly predict rating

Conclusion, Future Scope & Contributions

Conclusion

This project analysed 23,000+ BigBasket product listings across 10 structured Python analyses — from descriptive statistics and distribution analysis to Pearson correlation, brand rating benchmarking, and price segment deep-dives — to answer a core business question: **what drives pricing, discounts, and customer satisfaction?**

The central finding is that **price does not drive customer satisfaction**. The correlation between sale price and rating is very weak; deep discounts are associated with lower, not higher, ratings. What matters is product quality, brand reputation, and category membership — factors that pricing levers alone cannot fix.



The three highest-impact actions: (1) auditing and delisting deeply discounted, low-rated SKUs; (2) promoting mid-range, high-rating brands in marketing; and (3) improving product descriptions in low-performing sub-categories — require no new data and can be actioned immediately.

Future Scope

→

Time-Series Analysis

Historical snapshots would reveal whether quality is improving or declining by category.

→

Predictive Rating Model

With description length, image count, and sub-category features, an ML model could predict a new product's likely rating before launch.

→

Sales Volume Integration

Transaction data would enable sales-weighted analysis, giving high-selling products proportionally more influence.

→

Competitor Benchmarking

Comparable data from Blinkit/JioMart would enable market-level pricing and assortment gap analysis.

Notebook Map

Notebook	Purpose
01_data_overview.ipynb	Initial data inspection — shape, dtypes, missing values
02_cleaning.ipynb	Full ETL pipeline — cleaning steps and feature engineering
03_eda.ipynb	10 analyses — all EDA and statistical tests
04_statistical_analysis.ipynb	Statistical tests — correlation, ANOVA, regression
05_final_load_prep.ipynb	KPI computation and Tableau-ready CSV export

Contribution Matrix

Team Member	Contribution
Amiti Aneesh	Data ingestion, Tableau dashboard design (01_extraction.ipynb, 05_final_load_prep.ipynb)
Lakshith	Trend analysis, statistical tests, KPI computation (04_statistical_analysis.ipynb)
Chanakya Shinde	EDA, ETL pipeline, feature engineering (02_cleaning.ipynb, 03_eda.ipynb)
Sai Praneeth Sharma	Business insights, recommendations, final report & presentation